

Enhanced Volcanic Signal Detection in Santorini: Implications for Sustainable Tourism Development

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(Marsellos & Kidd, 2008)

Santorini

Volcanic Arc in South Aegean



Introduction

Santorini: An iconic super-volcano island in the Aegean Sea, Europe.

- Importance of monitoring volcanic activity for tourism and local safety.
- Advanced time series analysis for detecting volcanic thermal anomalies.



NSF-Geoteam in action installing a new thermo-probe on the Santorini Volcano



Data collection and methodology

Ground Temperature Data on fumaroles:

Collected from volcanic sites and meteorological stations.

Ground thermometer at 35 cm depth.

Wavelet Analysis:

Identifies temperature anomalies across timescales.

Kolmogorov-Zurbenko Filter:

Separates atmospheric effects from volcanic signals.

Introduction - fumaroles

(Left) Pointing with a green glove to an active sulfuric fumarole at Stefanos Crater (Nisyros) emitting gas at 99 °C. (Right) Faith Renner next to a fumarole inside Stefanos Crater.

A



B



C



Methods: Continuous Thermal analysis

Volcanic temperature hourly data originated from active craters, e.g.:

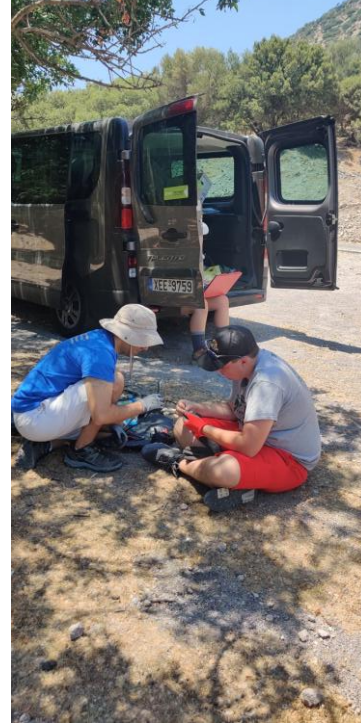
- **At Stephanos crater on Nisyros volcanic island**
- **Using an in-situ temperature dual probe Tinytag TGP-4520**



Thermal scanning and identification of best locations for continuous temperature monitoring

Hofstra students programming sensors and drone flight for installation at the active volcano of Nisyros

Hofstra students programming sensors, loggers, and drone flight for installation at the active volcano



Fumaroles continuous thermal monitoring

Nick Ptak and Faith Renner (students) deploy the ground thermometers on the active crater



Fumaroles continuous thermal monitoring

Nick Ptak and Faith Renner
deploying the temperature
loggers on the active crater



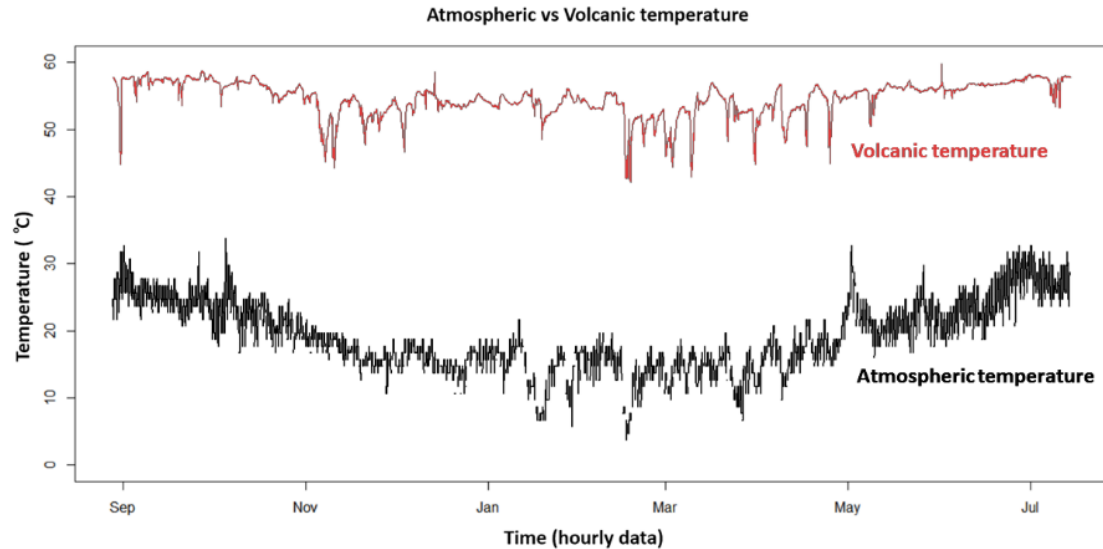
Fumaroles continuous thermal monitoring

Faith Renner accessing an active fumarole to install a ground thermometer inside the crater



Data collection and methodology

- Raw temperature data from weather station and ground thermometer at Santorini volcano (Nea Kameni).
- Challenge to distinguish between atmospheric and real volcanic temperature variations.



Methodology

Wavelet Analysis:

- Utilized WaveletComp package in R.
- Identifies distinct patterns and oscillations in temperature data.

Kolmogorov-Zurbenko Filter:

- Facilitates decomposition of temperature data into defined segments.
- Highlights temporal components from long-term to short-term fluctuations.

Kolmogorov - Zurbenko filter (KZ)

$$Y_t = \frac{1}{m} \sum_{j=-k}^k (X_{t+j})$$

- **Why KZ moving average?**
- KZ-Filter equation utilized for the moving average adapted from Zurbenko (1986); package kza (Close et al. 2020).

Function:

- Reduces short-term fluctuations and outliers.
- Enhances pattern and frequency identification.

Implementation:

- Applied with minimal window size and few iterations.

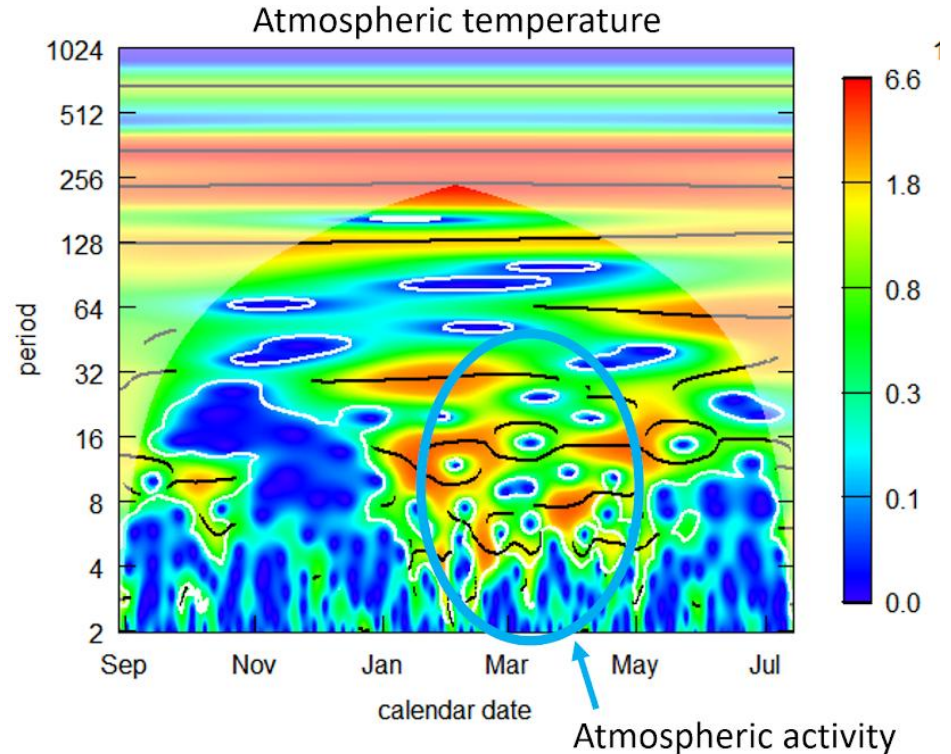
Results

Wavelet Analysis Findings:

- Temperature fluctuations indicating volcanic activity.
- Separation of atmospheric and volcanic thermal signals.

Significance:

- Identification of periodic temperature shifts as potential volcanic precursors.



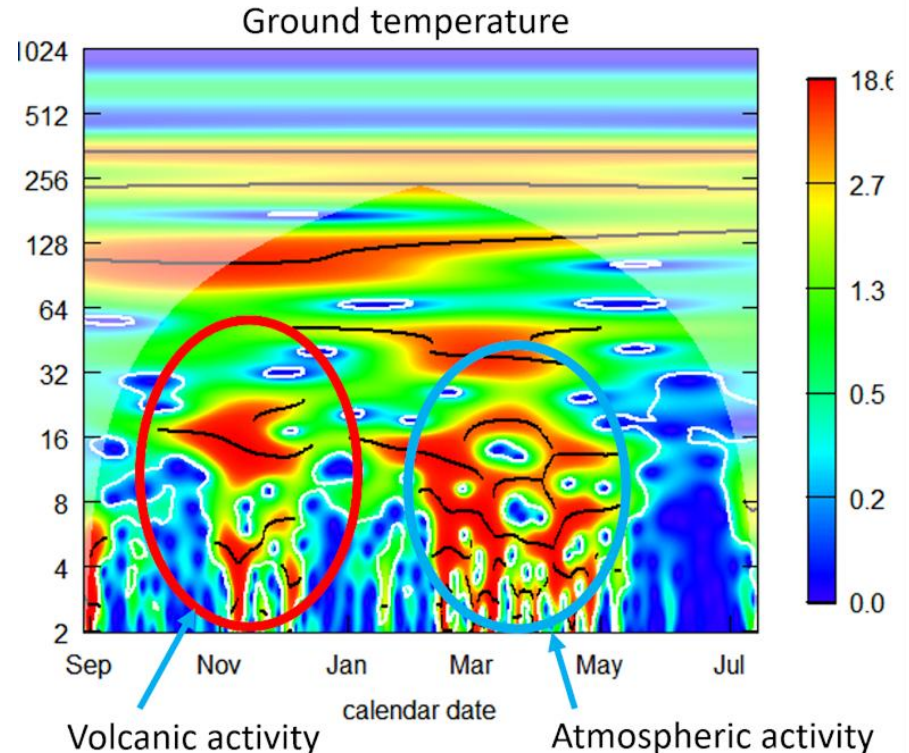
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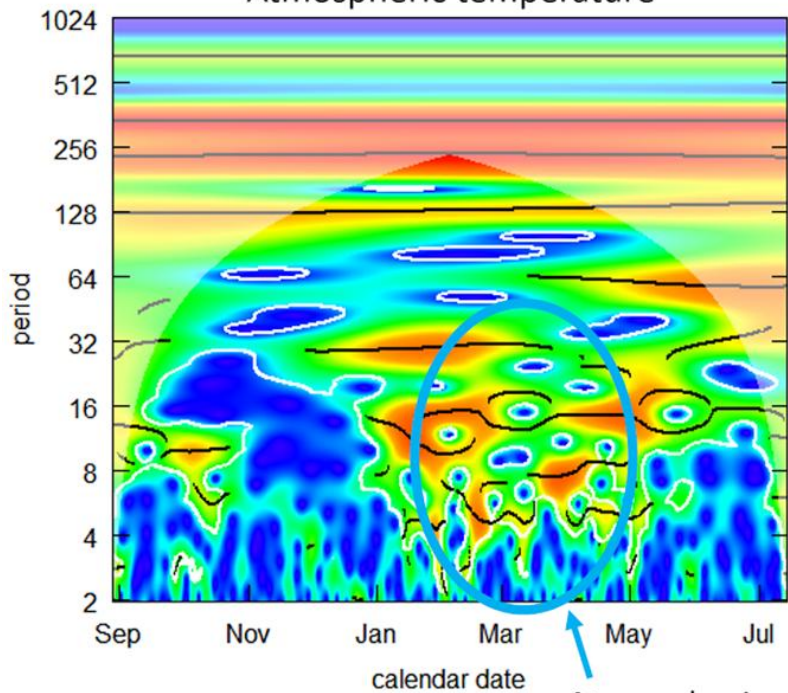
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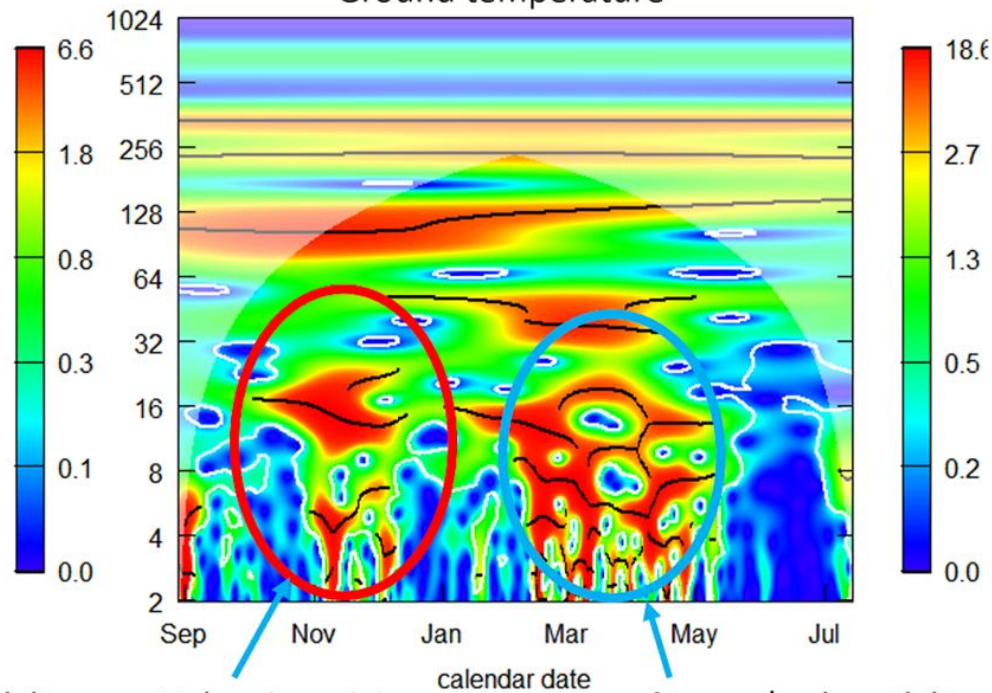
Results

Atmospheric temperature



Atmospheric activity

Ground temperature



Volcanic activity

Atmospheric activity

Discussion

Analytical Insights:

- Balance between frequency resolution and time resolution.
- Importance of high-frequency seasonalities for volcanic monitoring.

Challenges:

- Selecting between finer frequency resolution and precise time resolution.

Implications:

- Enhanced signal detection for timely volcanic activity prediction.

Conclusion

Key Findings:

- Improved detection of volcanic signals using wavelet analysis and Kolmogorov-Zurbenko filter.
- Higher-frequency seasonalities in hourly volcanic temperature data.

Recommendations:

- Ongoing refinement of analytical techniques.
- Development of effective early warning systems for sustainable tourism.

References

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